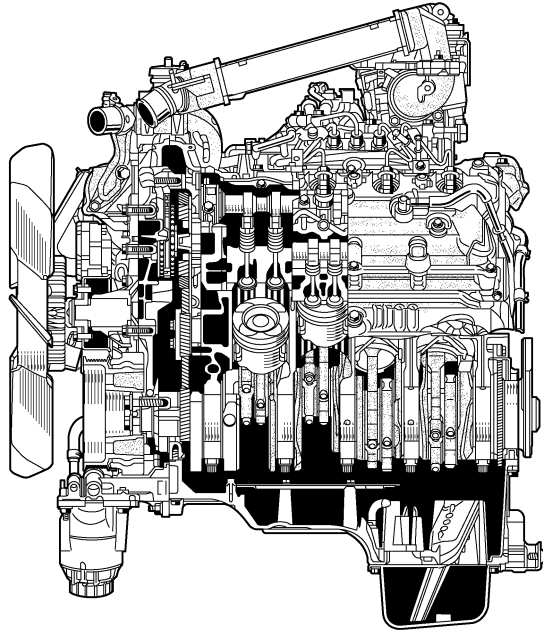


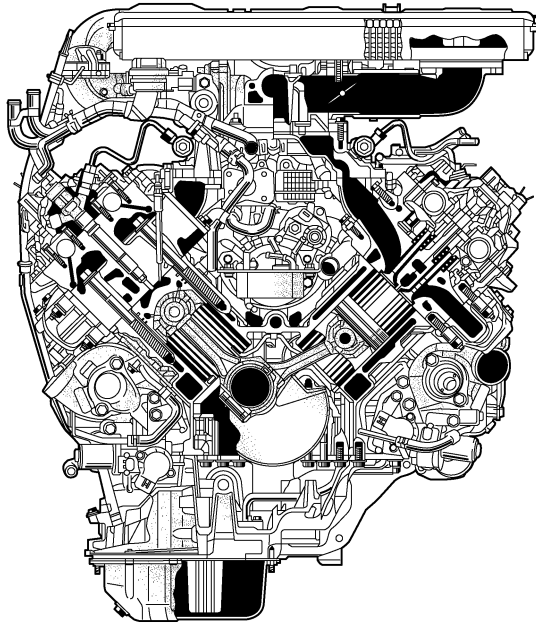
1VD-FTV ENGINE

■ DESCRIPTION

The 1VD-FTV engine is newly provided. The 1VD-FTV TOYOTA D-4D (Direct injection 4 stroke common-rail Diesel engine) is V8 4.5 liter, 32-valve DOHC, turbocharged diesel engine with intercooler. This engine uses direct injection system and common-rail system to realize high performance, clean emission, low noise and low vibration.



081EG04Y



081EG05Y

► Engine Specifications ◀

No. of Cyls. & Arrangement			8-Cylinder, V Type	
Valve Mechanism			32-valve, DOHC, Chain & Gear Drive	
Combustion Chamber			Direct Injection Type	
Manifolds			Cross-flow	
Fuel System			Common-rail Type	
Displacement		cm ³ (cu. in.)	4461 (272.2)	
Bore × Stroke		mm (in.)	86.0 × 96.0 (3.39 × 3.78)	
Compression Ratio			16.8 : 1	
Max. Output [ECE]	Europe		210 kW @ 3600 rpm	
	Australia and Europe (Models with pre-cleaner)		195 kW @ 3400 rpm	
	G.C.C. Countries and General Countries	A/T	173 kW @ 3200 rpm	
		M/T	162 kW @ 3600 rpm	
Max. Torque [ECE]	Europe		650 N·m @ 1600 to 2800 rpm	
	Australia and Europe (Models with pre-cleaner)		650 N·m @ 1600 to 2600 rpm	
	G.C.C. Countries and General Countries	A/T	615 N·m @ 1800 to 2200 rpm	
		M/T	430 N·m @ 1200 to 3600 rpm	
Valve Timing	Intake	Open	2° BTDC	
		Close	17° ABDC	
	Exhaust	Open	51° BBDC	
		Close	2° ATDC	
Firing Order			1-2-7-3-4-5-6-8	
Oil Grade	Europe		ACEA B1, API CF-4 or CF	
	Except Europe		G-DLD-1, API CF-4 or CF	
Fuel Cetane Number	Europe		48 or higher	
	Except Europe		50 (Cetane Index 45) or higher	
Emission Regulation	Europe	EURO IV		
		EURO III*1		
	Australia	EURO IV		
	G.C.C. Countries	—		
	General Countries	EURO III*2		
		EURO II*3		
—				
Engine Service Mass*4 (Reference) kg (lb)	Models with EGR	A/T	325 (717)	
		M/T	340 (750)	
	Models without EGR	A/T	320 (705)	
		M/T	334 (736)	

*1: Russian package models

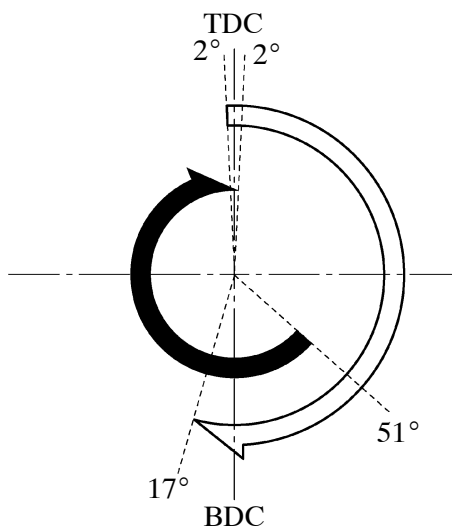
*2: Package models for South Africa, Argentina and Papua New Guinea

*3: Package models for Philippines and Vietnam

*4: The figure shown is the weight of the part without coolant and oil.

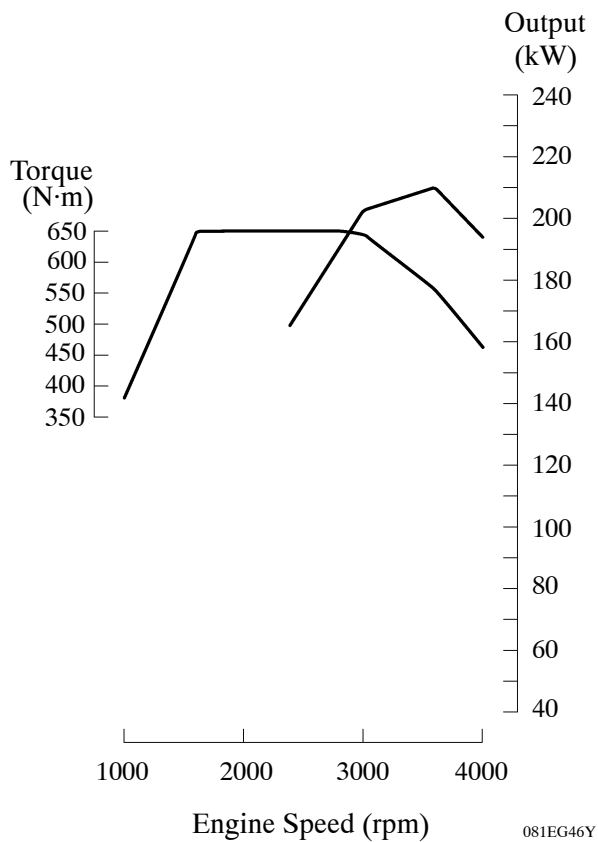
► Valve Timing ◀

↗ : Intake Valve Opening Angle
↘ : Exhaust Valve Opening Angle

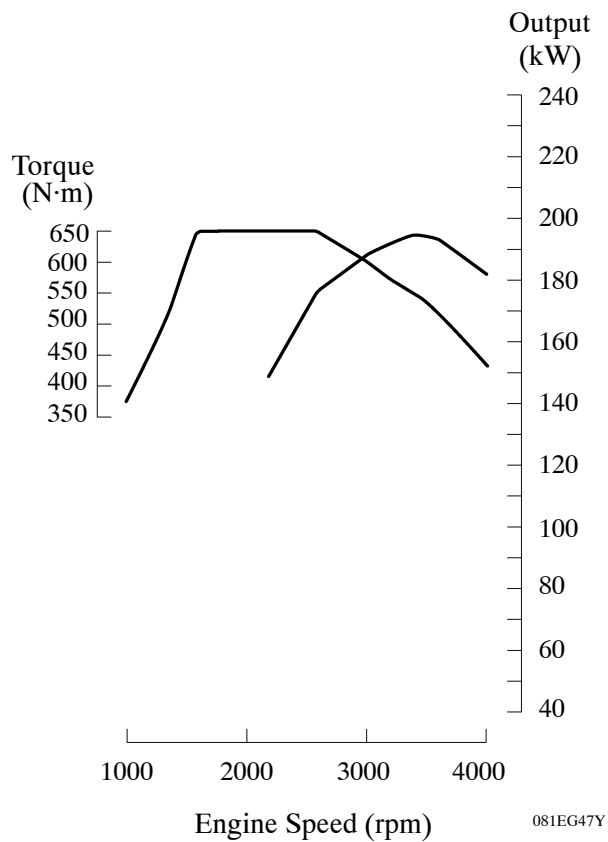


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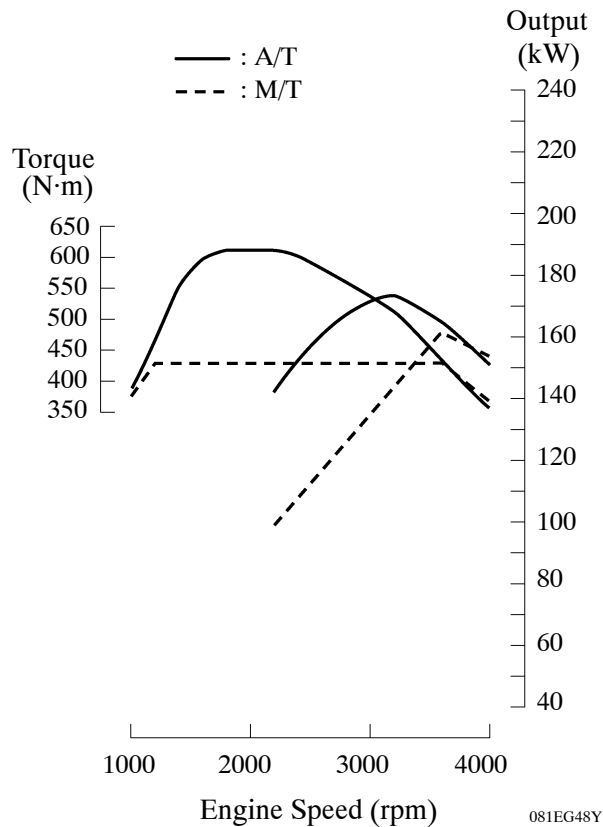
► Performance Curve ◀



Europe



Australia and
Europe (Models with pre-cleaner)



G.C.C. Countries and General Countries